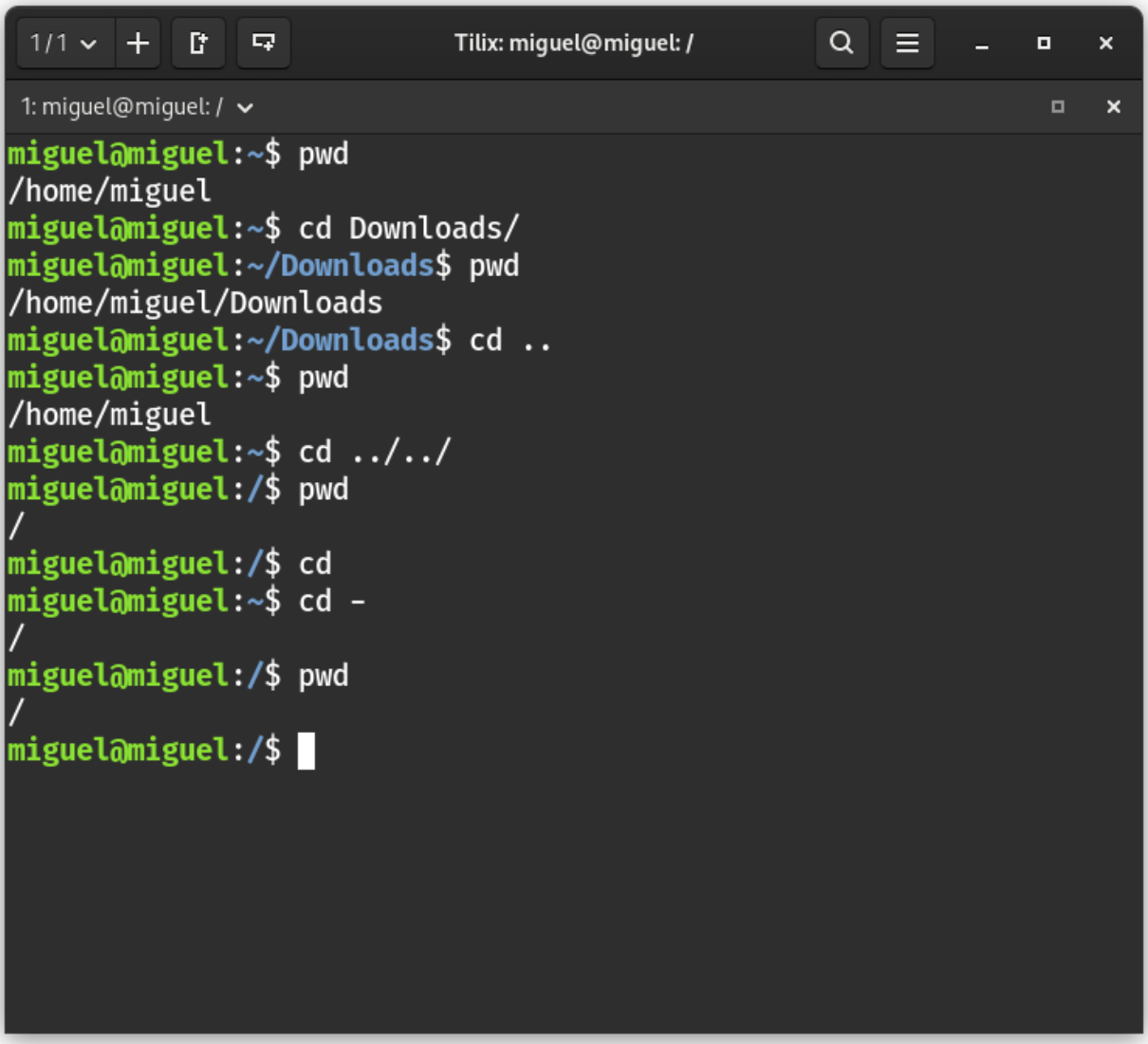
Week report 5

The Linux Filesystem

- **Practice 1:**



```
miguel@miguel:~$ wget https://cis106.com/assets/practicels.tar.xz
--2026-03-25 15:12:21--  https://cis106.com/assets/practicels.tar
Resolving cis106.com (cis106.com)... 185.199.108.153
Connecting to cis106.com (cis106.com)|185.199.108.153|:443... con
HTTP request sent, awaiting response... 200 OK
Length: 2229212 (2.1M) [application/x-xz]
Saving to: 'practicels.tar.xz'

practicels.tar.x 100%[========>]   2.12M  --.-KB/s    in 0.09s

2026-03-25 15:12:21 (23.8 MB/s) - 'practicels.tar.xz' saved [2229

miguel@miguel:~$ tar Jxf practicels.tar.xz
miguel@miguel:~$ cd practicels/
miguel@miguel:~/practicels$ ls
book.txt            homework.docx  song.mp3
countries.xml       movie.mp4      tux-nobg.png
history-debian.pdf  people.csv     tux-penguin.gif
miguel@miguel:~/practicels$ ls -a
.          countries.xml      movie.mp4  tux-nobg.png
..         history-debian.pdf  people.csv  tux-penguin.gif
book.txt   homework.docx      song.mp3
miguel@miguel:~/practicels$ ls -l
total 2932
-rw-rw-r-- 1 miguel miguel    1043 Feb 27  2022 book.txt
-rw-rw-r-- 1 miguel miguel   23972 Feb 27  2022 countries.xml
-rw-rw-r-- 1 miguel miguel  163057 Feb 27  2022 history-debian.pd
-rw-rw-r-- 1 miguel miguel  111303 Feb 22  2022 homework.docx
-rw-rw-r-- 1 miguel miguel 1570024 Feb 27  2022 movie.mp4
-rw-rw-r-- 1 miguel miguel  284042 Feb 27  2022 people.csv
-rw-rw-r-- 1 miguel miguel  764176 Feb 27  2022 song.mp3
-rw-rw-r-- 1 miguel miguel   21314 Feb 27  2022 tux-nobg.png
-rw-rw-r-- 1 miguel miguel   44299 Jan 31  2020 tux-penguin.gif
miguel@miguel:~/practicels$ ls -lh
total 2.9M
-rw-rw-r-- 1 miguel miguel 1.1K Feb 27  2022 book.txt
-rw-rw-r-- 1 miguel miguel  24K Feb 27  2022 countries.xml
-rw-rw-r-- 1 miguel miguel 160K Feb 27  2022 history-debian.pdf
-rw-rw-r-- 1 miguel miguel 109K Feb 22  2022 homework.docx
-rw-rw-r-- 1 miguel miguel 1.5M Feb 27  2022 movie.mp4
-rw-rw-r-- 1 miguel miguel 278K Feb 27  2022 people.csv
-rw-rw-r-- 1 miguel miguel 747K Feb 27  2022 song.mp3
-rw-rw-r-- 1 miguel miguel  21K Feb 27  2022 tux-nobg.png
-rw-rw-r-- 1 miguel miguel  44K Jan 31  2020 tux-penguin.gif
```

- **Practice 2:**

```
miguel@miguel:~/practicels$ ls -lht
total 2.9M
-rw-rw-r-- 1 miguel miguel  24K Feb 27  2022 countries.xml
-rw-rw-r-- 1 miguel miguel 278K Feb 27  2022 people.csv
-rw-rw-r-- 1 miguel miguel 747K Feb 27  2022 song.mp3
-rw-rw-r-- 1 miguel miguel 1.5M Feb 27  2022 movie.mp4
-rw-rw-r-- 1 miguel miguel 160K Feb 27  2022 history-debian.pdf
-rw-rw-r-- 1 miguel miguel  21K Feb 27  2022 tux-nobg.png
-rw-rw-r-- 1 miguel miguel 1.1K Feb 27  2022 book.txt
-rw-rw-r-- 1 miguel miguel 109K Feb 22  2022 homework.docx
-rw-rw-r-- 1 miguel miguel  44K Jan 31  2020 tux-penguin.gif
miguel@miguel:~/practicels$ ls -lhgGs
total 2.9M
4.0K -rw-rw-r-- 1 1.1K Feb 27  2022 book.txt
 24K -rw-rw-r-- 1  24K Feb 27  2022 countries.xml
160K -rw-rw-r-- 1 160K Feb 27  2022 history-debian.pdf
112K -rw-rw-r-- 1 109K Feb 22  2022 homework.docx
1.5M -rw-rw-r-- 1 1.5M Feb 27  2022 movie.mp4
280K -rw-rw-r-- 1 278K Feb 27  2022 people.csv
748K -rw-rw-r-- 1 747K Feb 27  2022 song.mp3
 24K -rw-rw-r-- 1  21K Feb 27  2022 tux-nobg.png
 44K -rw-rw-r-- 1  44K Jan 31  2020 tux-penguin.gif
miguel@miguel:~/practicels$ ls -mr
tux-penguin.gif, tux-nobg.png, song.mp3, people.csv, movie.mp4,
homework.docx, history-debian.pdf, countries.xml, book.txt
miguel@miguel:~/practicels$ ls -1X
people.csv
homework.docx
tux-penguin.gif
song.mp3
movie.mp4
history-debian.pdf
tux-nobg.png
book.txt
countries.xml
```

- **Practice 3:**

```
miguel@miguel:~$ pwd
/home/miguel
miguel@miguel:~$ cd
miguel@miguel:~$ cd /
miguel@miguel:/$ cd usr/share/themes
miguel@miguel:/usr/share/themes$ cd ../../
miguel@miguel:/usr$ ls -A ~
.bash_aliases    .fonts             Public
.bash_history    .gitconfig         Scripts
.bash_logout     .gnupg             snap
.bashrc          lab5files          .ssh
.cache           lab5files.zip      Templates
cis106           .local             .vboxclient-clipboard-tty2-control.pid
.config          marstenHouse       .vboxclient-clipboard-tty2-service.pid
Desktop          marstenhouse.sh    .vboxclient-draganddrop-tty2-control.pid
Documents        .mozilla           .vboxclient-hostversion-tty2-control.pid
.dotnet          Music              .vboxclient-seamless-tty2-control.pid
Downloads        .nvm               .vboxclient-vmsvga-session-tty2-control.pid
essentials.sh    Pictures           .vboxclient-vmsvga-session-tty2-service.pid
essentials.sh.1  .pki               Videos
.face            practicels         .vscode
.face.icon       practicels.tar.xz  vscode.sh
figlet-fonts     .profile           .wget-hsts
miguel@miguel:/usr$ cd; ls -tX https://github.com/miguellopez-cis106/cis106/
ls: cannot access 'https://github.com/miguellopez-cis106/cis106/': No such file or directory
miguel@miguel:~$ cd; ls -tX cis106/
cheats  week_reports            ChallengeQuestion_v2.png
labs    bashversion.png         lookingfordifferentterminals.png
notes   ChallangeQuestion.png   playing_with_commands.png
```

```
miguel@miguel:~$ ls -rR ~/cis106/
/home/miguel/cis106/:
week_reports                    cheats
playing_with_commands.png       ChallengeQuestion_v2.png
notes                           ChallangeQuestion.png
lookingfordifferentterminals.png bashversion.png
labs

/home/miguel/cis106/week_reports:
wr5  wr4  wr3  wr2  wr1

/home/miguel/cis106/week_reports/wr5:
wr5.md  practice2.png  practice2.2.png  practice1.png

/home/miguel/cis106/week_reports/wr4:
wr4.pdf  wr4.md  practice2.png  practice1.png  hello.sh

/home/miguel/cis106/week_reports/wr3:
wr3.pdf  practice9.png  practice7.png  practice5.png  practice3.png
wr3.md   practice8.png  practice6.png  practice4.png  practice10.png

/home/miguel/cis106/week_reports/wr2:
wr2.pdf  wr2.md  debianDesktop.png

/home/miguel/cis106/week_reports/wr1:
wr1.pdf  wr1.md  slack.png.png  github.png.png

/home/miguel/cis106/notes:
notes5  notes4  notes3  notes2  notes1

/home/miguel/cis106/notes/notes5:
usingcd.png  notes5.pdf  notes5.md  lsexample.png

/home/miguel/cis106/notes/notes4:
 shellscript.png  'Screenshot from 2026-03-08 23-42-20.png'   notes4.pdf
 shellcript.sh     runningthescript.png                       notes4.md

/home/miguel/cis106/notes/notes3:
notes3.pdf  notes3.md

/home/miguel/cis106/notes/notes2:
notes2.pdf  notes2.md
```

```
miguel@miguel:~$ ls -lh ~
total 5.7M
drwxrwxr-x 7 miguel miguel 4.0K Feb 12 18:41 cis106
drwxr-xr-x 2 miguel miguel 4.0K Feb 12 00:17 Desktop
drwxr-xr-x 4 miguel miguel 4.0K Mar 25 00:15 Documents
drwxr-xr-x 3 miguel miguel 4.0K Mar 19 18:02 Downloads
-rwxrwxr-x 1 miguel miguel 7.6K Jan 22 11:04 essentials.sh
-rw-rw-r-- 1 miguel miguel 7.6K Jan 22 11:04 essentials.sh.1
drwxrwxr-x 3 miguel miguel  20K Mar 13 22:49 figlet-fonts
drwxrwxr-x 2 miguel miguel 4.0K Feb 22  2022 lab5files
-rw-rw-r-- 1 miguel miguel 3.5M Mar  6 13:38 lab5files.zip
drwxrwxr-x 4 miguel miguel 4.0K Mar 18 23:38 marstenHouse
-rw-rw-r-- 1 miguel miguel 3.4K Mar  6 13:38 marstenhouse.sh
drwxr-xr-x 2 miguel miguel 4.0K Feb 12 00:17 Music
drwxr-xr-x 3 miguel miguel 4.0K Mar 19 17:52 Pictures
drwxrwxr-x 2 miguel miguel 4.0K Feb 27  2022 practicels
-rw-rw-r-- 1 miguel miguel 2.2M Mar 19 13:48 practicels.tar.xz
drwxr-xr-x 2 miguel miguel 4.0K Feb 12 00:17 Public
drwxrwxr-x 2 miguel miguel 4.0K Mar 13 23:15 Scripts
drwx------ 3 miguel miguel 4.0K Feb 22 00:54 snap
drwxr-xr-x 2 miguel miguel 4.0K Feb 12 00:17 Templates
drwxr-xr-x 3 miguel miguel 4.0K Mar 12 18:26 Videos
-rwxrwxr-x 1 miguel miguel 2.5K Jan 22 11:04 vscode.sh
```

```
miguel@miguel:~/cis106$ ls -lgG ~
total 5768
drwxrwxr-x 7    4096 Feb 12 18:41 cis106
drwxr-xr-x 2    4096 Feb 12 00:17 Desktop
drwxr-xr-x 4    4096 Mar 25 00:15 Documents
drwxr-xr-x 3    4096 Mar 19 18:02 Downloads
-rwxrwxr-x 1    7747 Jan 22 11:04 essentials.sh
-rw-rw-r-- 1    7747 Jan 22 11:04 essentials.sh.1
drwxrwxr-x 3   20480 Mar 13 22:49 figlet-fonts
drwxrwxr-x 2    4096 Feb 22  2022 lab5files
-rw-rw-r-- 1 3570561 Mar  6 13:38 lab5files.zip
drwxrwxr-x 4    4096 Mar 18 23:38 marstenHouse
-rw-rw-r-- 1    3407 Mar  6 13:38 marstenhouse.sh
drwxr-xr-x 2    4096 Feb 12 00:17 Music
drwxr-xr-x 3    4096 Mar 19 17:52 Pictures
drwxrwxr-x 2    4096 Feb 27  2022 practicels
-rw-rw-r-- 1 2229212 Mar 19 13:48 practicels.tar.xz
drwxr-xr-x 2    4096 Feb 12 00:17 Public
drwxrwxr-x 2    4096 Mar 13 23:15 Scripts
drwx------ 3    4096 Feb 22 00:54 snap
drwxr-xr-x 2    4096 Feb 12 00:17 Templates
drwxr-xr-x 3    4096 Mar 12 18:26 Videos
-rwxrwxr-x 1    2501 Jan 22 11:04 vscode.sh
miguel@miguel:~/cis106$
```

Shell Scripting Chapter 2

- **Practice 2:**

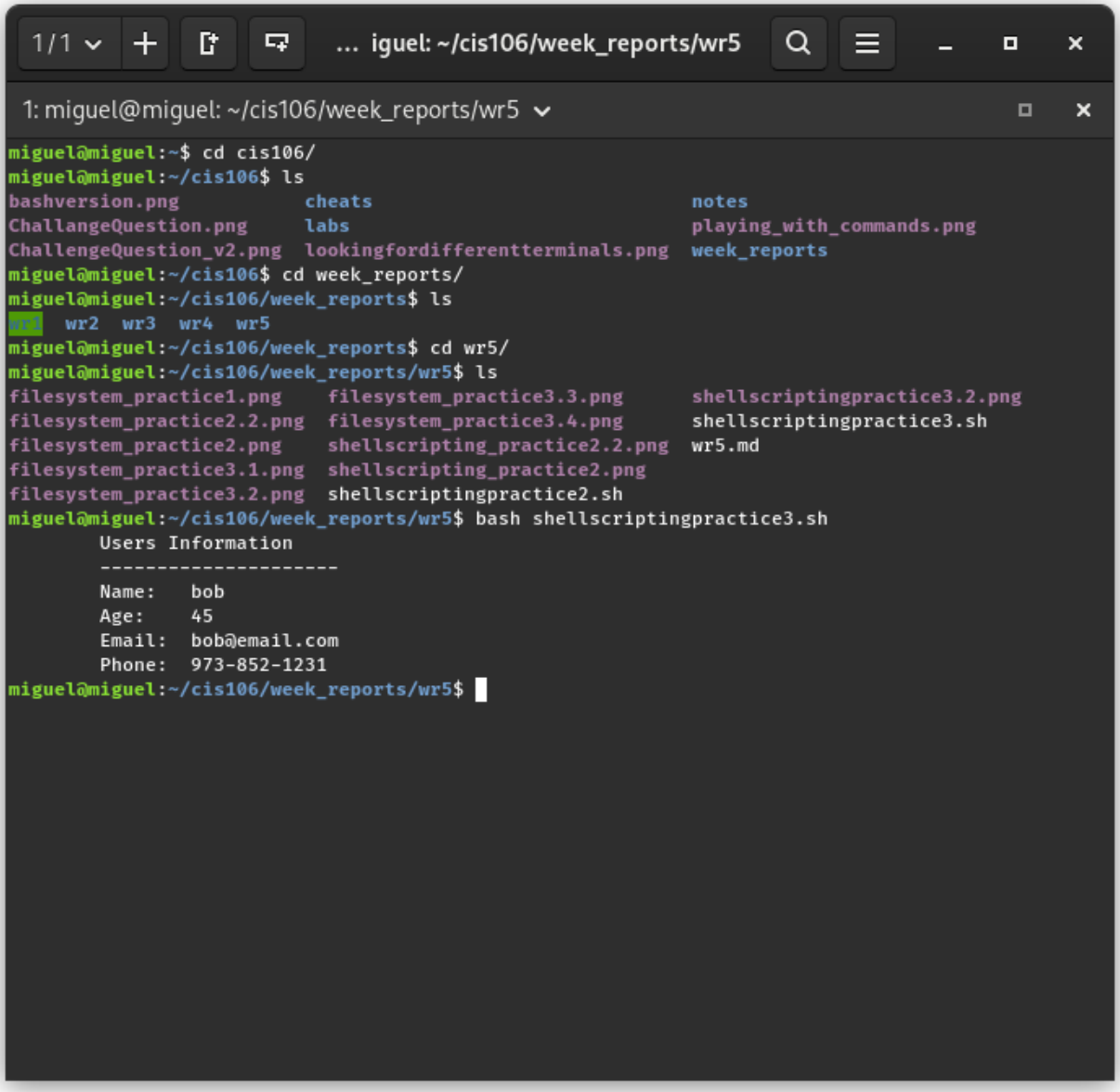
```
miguel@miguel:~/cis106/week_reports/wr5$ bash shellscriptingpractice2.sh
The Current User: miguel
The Current Shell: /bin/bash
The Current's User Home: /home/miguel
The Path Var:
 /home/miguel/.nvm/versions/node/v24.13.1/bin:/usr/local/bin:/usr/bin:/bin:/usr/local/games:/usr
/games:/snap/bin:/usr/bin
miguel@miguel:~/cis106/week_reports/wr5$
```

Open

shellscriptingpractice2.sh
~/cis106/week_reports/wr5

```
#!/bin/bash
echo "The Current User: $USER"
echo "The Current Shell: $SHELL"
echo "The Current's User Home: $HOME"
echo -e "The Path Var: \n $PATH"
```

- **Practice 3:**


1/1
... iguel: ~/cis106/week_reports/wr5
1: miguel@miguel: ~/cis106/week_reports/wr5
miguel@miguel:~$ cd cis106/
miguel@miguel:~/cis106$ ls
bashversion.png cheats notes
ChallangeQuestion.png labs playing_with_commands.png
ChallengeQuestion_v2.png lookingfordifferentterminals.png week_reports
miguel@miguel:~/cis106$ cd week_reports/
miguel@miguel:~/cis106/week_reports$ ls
wr1 wr2 wr3 wr4 wr5
miguel@miguel:~/cis106/week_reports$ cd wr5/
miguel@miguel:~/cis106/week_reports/wr5$ ls
filesystem_practice1.png filesystem_practice3.3.png shellscriptingpractice3.2.png
filesystem_practice2.2.png filesystem_practice3.4.png shellscriptingpractice3.sh
filesystem_practice2.png shellscripting_practice2.2.png wr5.md
filesystem_practice3.1.png shellscripting_practice2.png
filesystem_practice3.2.png shellscriptingpractice2.sh
miguel@miguel:~/cis106/week_reports/wr5$ bash shellscriptingpractice3.sh
Users Information

Name: bob
Age: 45
Email: bob@email.com
Phone: 973-852-1231
miguel@miguel:~/cis106/week_reports/wr5$

Open

shellscriptingpractice3.sh
~/cis106/week_reports/wr5

```bash
#!/bin/bash/
# Variables that store age, name, email, and phone number
name="bob"
age=45
email="bob@email.com"
phone="973-852-1231"

# Display the users info
echo -e "\tUsers Information"
echo -e "\t———————————————————"
echo -e "\tName: \t$name" #added a tab for presentation
echo -e "\tAge: \t$age"
echo -e "\tEmail: \t$email"
echo -e "\tPhone: \t$phone"
```